HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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Abstract. A Hamiltonian bundle $M \hookrightarrow P \to X$ (with monotone compact fibers) induces a kind of "bundle of A_∞ categories" over X, with fiber given by the Fukaya category of M. Algebraic K-theory ideas, the above picture for $X = S^m$, and basic representation theory yield the following: if G is a compact Lie group and R is any commutative ring then there is a natural group homomorphism $\pi_m(BG) \to K_m^{\mathbb{Z}_2}(R)$. Here $K_m^{\mathbb{Z}_2}(R)$ is a variant of algebraic K-theory of R, isomorphic to $K_m(R) \oplus K_{m-1}(R)$ if R is a regular ring (e.g. $\mathbb{Z}$). Our construction naturally first assigns elements in a certain categorified algebraic K-theory $K_m^{Cat}(R)$. And the construction yields that $K_2^{Cat}(\mathbb{Z})$ is infinitely generated (with details postponed). This is in contrast to finite generation of standard algebraic K-theory of $\mathbb{Z}$ already proved by Quillen. This categorified algebraic K-theory is analogous to Toën's secondary K-theory of R. Taking the Langlands dual of G, we explore a conjectural relationship between the images of the corresponding homomorphisms above.

1. Introduction

A guiding idea is that a Hamiltonian bundle $M \hookrightarrow P \to Y$ gives rise to a kind of "bundle of A_∞ categories", with fiber the Fukaya category of M. Precisely what this means is an interesting mathematical problem in its own right, and one answer is developed in [17]. Note that we are not talking about local systems, as the "bundles" that one gets are generally non-trivial over spheres as shown in [16].

The second idea, due to Toën [21], is that there is a refinement of K-theory of a commutative ring R, by replacing R-modules with dg or A_∞ categories over R, and using Waldhausen K-theory of a commutative ring. We show that these two ideas naturally fit together. We use the above assignment, with $Y = S^m$, to associate to P elements in categorified and standard algebraic K-theory. This culminates in a connection between symplectic geometry and algebraic K-theory, and in particular we get a certain geometric insight into what we call categorified algebraic K-theory of $\mathbb{Z}$. For example, we obtain an infinite generation result. At the end we explore connections with Langlands duality.

Although in the main examples here the principal actors are compact groups, in particular $PU(n)$, the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds; so we quickly introduce this. Let (M, ω) be a symplectic manifold, throughout assumed to be compact, with ω a closed nondegenerate 2-form on M. The group of diffeomorphisms ϕ of M, satisfying $\phi^*\omega = \omega$, is called the symplectomorphism group of (M, ω). When M is simply connected, this is also the group of Hamiltonian symplectomorphisms, which we denote by $\mathcal{H} = \mathrm{Ham}(M, \omega)$. In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields $\{X_t\}$, $t \in [0, 1]$, satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions $\{H_t\}$ on M.

The group $\mathcal{H}$ is one of the more enigmatic objects of mathematics. It is always (unless $M = pt$) an infinite-dimensional Fréchet Lie group with its C^∞ topology. But it has a natural, bi-invariant

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Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics, see for instance Polterovich [12].

Much of the algebraic/topological structure of $\mathcal{H}$ is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration* $M \hookrightarrow P \to Y$ is a smooth fiber bundle over Y, with fiber a symplectic manifold (M, ω), whose structure group is contained in $\mathrm{Ham}(M, \omega)$. For example, take $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$ where ω_{FS} is the Fubini-Study symplectic form, and take the structure group $G = PU(n+1) \subset \mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$.

The other ingredient is algebraic K-theory, which is a construction associating an infinite loop space [1] $K(R)$ to any ring R , and so abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called K-theory groups of R. This is a powerful tool in algebraic topology and ring theory, with applications for instance in number theory and geometric topology.

One motivating example for algebraic K-theory is the following. Take R to be the non-commutative ring $\mathcal{K}$ of compact operators on a separable complex Hilbert space, the algebraic K-theory space $K(\mathcal{K})$ [2] is identified with the topological K-theory space $KU = BU \times \mathbb{Z}$, see Karoubi [6], Suslin-Wodzicki [20]. In particular, a complex vector bundle E over a space X determines a homotopy class $[f_E : X \to K(\mathcal{K})]$, which we might call a *geometric element* of the corresponding abelian group. We shall see that in a certain sense this extends to a general ring R, provided it is commutative, and replacing complex vector bundles by Hamiltonian fibrations.

For each commutative ring R, we construct "categorified algebraic K-theory groups" $K_m^{Cat}(R)$. Actually it will be convenient to define more general groups $K^{Cat}(X, R)$ for each Kan complex X, generated by aforementioned bundles of A_∞ categories over X with fibers A_∞ categories with finite type, see Definition 3.1. This is analogous to the group of classes $[Y \to K(\mathcal{K})]$ for a space Y, i.e. the topological K-theory of Y.

In principle, this categorified algebraic K-theory of R is a slight variation of the secondary K-theory of R introduced by Toën, with the main difference that $\mathbb{Z}$-graded chain complexes are replaced by $\mathbb{Z}_2$-graded chain complexes. By way of the Hochschild chain functor, $K_m^{Cat}(R)$ should admit a natural map to the 2-periodic variant of algebraic K-theory of R, which we denote by $K_m^{\mathbb{Z}_2}(R)$, see Remark 3.6, although we don't yet need this.

Remark 1.1. It was pointed out to me by Bertrand Toën that for R a regular ring, for example $R = \mathbb{Z}$, $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$. This can be proved with A^1 homotopy theory.

To obtain "geometric elements", we start with a Hamiltonian fibration $M \hookrightarrow P \to Y$, with fiber a monotone symplectic manifold. The following is a conjunction of Theorem 4.5 and Corollary 4.8.

Theorem 1.2. *If the Fukaya category of M is finite type, then for each R the data of the isomorphism class of P determines an element in $K^{Cat}(Y_\bullet, R)$. In particular, this gives homomorphisms*

$$\phi_m : \pi_m(B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})) \to K_m^{Cat}(R),$$

and so mappings

$$\phi_m : \pi_m(BPU(n)) \to K_m^{Cat}(R).$$

Example 1. In the case of $n = 1$, in [16] using geometric arguments we outline the proof of non-triviality of ϕ_4 for $R = \mathbb{Z}$. This contrasts with a theorem of Rognes [14] on the triviality of $K_4(\mathbb{Z})$.

[1] In this paper, the term infinite loop space is meant to be synonymous with connective Ω-spectrum, i.e. a sequence of spaces $\{E_n\}_{n \in \mathbb{N}}$ with $E_n \simeq \Omega E_{n+1}$. Then $K(R)$ is the notation for the 0-space E_0 of the corresponding spectrum.

[2] Rather one must take a certain non-connective variant of the algebraic K theory construction.

Claim 1.3.
$$\phi_2 : \mathbb{Z}_p \simeq \pi_2(BPU(p)) \to K_2^{Cat}(\mathbb{Z})$$
is injective for each prime p. In particular, $K_2^{Cat}(\mathbb{Z})$ *is infinitely generated.*

Since the groups $K_m(\mathbb{Z})$ (and hence $K_m^{\mathbb{Z}_2}(\mathbb{Z})$ assuming Remark 1.1) are always finitely generated by the foundational results of Quillen [13], the claim is perhaps surprising.

The main geometric ingredients for the proof are the Seidel homomorphism [18] and the standard isomorphisms of the Hochschild homology of the Fukaya category with the quantum homology. The details will appear elsewhere.

Our construction also gives the following, which in principle lifts to categorified algebraic K-theory, see Section 4.1.

Theorem 1.4. *For any compact Lie group G, there is a homotopy natural map:*
$$h : \mathrm{BG} \to K^{\mathbb{Z}_2}(R), \tag{1.5}$$
and so there are natural homomorphisms
$$h_m : \pi_m(\mathrm{BG}) \to K_m^{\mathbb{Z}_2}(R).$$

Taking the Langlands dual of G, we explore correspondence of the respective mappings in Section 6.

Question 1. Can one obtain the classes
$$[\mathrm{BG} \to K^{\mathbb{Z}_2}(R)]$$
by techniques of homotopy theory and algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

It should be noted that compared to the maps ϕ_m, showing non triviality of the maps h_m is harder (and is an open problem), even restricting to $PU(n)$ and $R = \mathbb{Z}$. This might seem paradoxical, but the idea is that $K_m^{Cat}(R)$ sees more information, so it is easier to find nontrivial things in the image of ϕ_m.

2. Preliminaries

We will mostly follow Keller [7], but our base commutative ring will be denoted R to avoid overloading notation. An A_∞ category C over R is a form of non-associative dg-category, with $hom_C(a, b)$ a cochain complex over R. We refer the reader to standard references for further details, see for instance [11] which will come in use later.

Denote by $A_\infty \operatorname{Cat}_R^{strict}$ the category of strictly unital, $\mathbb{Z}$-graded A_∞ categories with morphisms strict unital A_∞ functors. We set $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$ to be like $A_\infty \operatorname{Cat}_R^{strict}$ but with chain complexes $\mathbb{Z}_n$-graded instead of $\mathbb{Z}$-graded. This means that all structure maps, functors, etc., are with respect to the $\mathbb{Z}_n$-grading. Various subscripts R may be omitted.

Let $\operatorname{Cat}_R^{\mathbb{Z}_n}$ denote the category of small R-linear $\mathbb{Z}_n$-graded categories. For $C \in \mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$, set $hC \in Cat_R^{\mathbb{Z}_n}$ to be the category with the same objects and with $hom_{hC}(a, b) = H^*(hom_C(a, b))$.

Given a morphism $f : A \to B$ in $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$, we denote by $f_* : hA \to hB$ the induced morphism in $Cat_R^{\mathbb{Z}_n}$. We say that f is a *quasi-equivalence* if f_* is an equivalence. We will say that f is a *Morita equivalence* if the induced functor $\mathcal{D}(B) \to \mathcal{D}(A)$ is an equivalence where $\mathcal{D}(A)$ denotes the derived category of A. We denote by $Hmo^{\mathbb{Z}_2}$ the homotopy category of $A_\infty \operatorname{Cat}_R^{strict,\mathbb{Z}_2}$ with respect to Morita equivalences, as explained by Keller this is a pointed category with the zero object: the category 0 with one object and zero endomorphism complex.

The category $\mathrm{A}_\infty\mathrm{Cat}_{\mathrm{R}}^{\mathbb{Z}_\mathrm{n},\mathrm{strict}}$ is cocomplete, see [11]. Furthermore, it has a model category structure [10]. This by left Bousfield localization yields a Morita model structure with weak equivalences Morita equivalences, although we do not explicitly need this model structure.

Definition 2.1. A ***short exact sequence*** in $A_\infty \operatorname{Cat}_R^{strict,\mathbb{Z}_2}$ is a equence $A \xrightarrow{I} B \xrightarrow{P} C$, such that in $Hmo^{\mathbb{Z}_2}$, I is the kernel of P and P is the cokernel of I.

As explained by Keller in [7], an explicit example for such a short exact sequence is given by a Drindfeld dg-quotient sequence $A \hookrightarrow B \to B/A$.

2.1. **Simplicial sets.** We will need basic theory of simplicial sets, particularly the notion of a simplex category, for the definition of bundles of A_∞ categories. We denote by Δ the simplex category:

- The set of objects of Δ is $\mathbb{N}$.
- $\hom_\Delta(n,m)$ is the set of non-decreasing maps $[n] \to [m]$, where $[n] = \{0, 1, \ldots, n\}$, with its natural order.

A simplicial set X is a functor
$$X : \Delta^{op} \to Set.$$
$X(n)$ is the set of n-simplices of X. Δ^d will denote a particular simplicial set: the standard representable d-simplex, with
$$\Delta^d(n) = hom_\Delta(n,d).$$
Morphisms of simplicial sets are given by natural trasnformations of functors and this forms a category denoted by $sSet$.

Definition 2.2. For X a simplicial set, $\Delta(X)$ will denote the ***simplex category of*** X. This is the category such that:

- The set of objects $\operatorname{obj}\Delta(X)$ is the set of simplicial maps:
$$\Sigma : \Delta^d \to X, \quad d \geq 0.$$
- Morphisms $f : \Sigma_1 \to \Sigma_2$ are commutative diagrams in $sSet$:

$$\begin{tikzcd} \Delta^d \arrow[r] \arrow[dr, "\Sigma_1"] & \Delta^n \arrow[d, "\Sigma_2"] \\ & X. \end{tikzcd} \tag{2.3}$$

Given a morphism $f : X \to Y$ of simplicial sets, we set $\Delta f : \Delta(X) \to \Delta(Y)$ to be the functor defined on objects by $\Delta f(\Sigma) = f \circ \Sigma$. If $m : \Sigma_1 \to \Sigma_2$ is a morphism in $\Delta(X)$:

$$\begin{tikzcd} \Delta^k \arrow[r, "m"] \arrow[dr, "\Sigma_1"] & \Delta^d \arrow[d, "\Sigma_2"] \\ & X, \end{tikzcd}$$

then obviously the diagram below also commutes:

$$\begin{tikzcd} \Delta^k \arrow[r, "m"] \arrow[dr, "h_1"] & \Delta^d \arrow[d, "h_2"] \\ & Y, \end{tikzcd}$$

where $h_i = \Delta f(\Sigma_i) = f \circ \Sigma_i$, $i = 1, 2$. And so the latter diagram determines a morphism $\Delta f(m) : h_1 \to h_2$ in $\Delta(Y)$. Clearly, this determines a functor Δf as needed.

Definition 2.4. Let X be a simplicial set, and C any category. Let $F_i : \Delta(X) \to C$, $i = 0, 1$ be functors. A ***concordance*** of F_0 to F_1 is a functor $\widetilde{F} : \Delta(X \times I) \to C$ such that $\widetilde{F}|_{\Delta(X\times\{0\})} = F_0$ and $\widetilde{F}|_{\Delta(X\times\{1\})} = F_1$.

3. Definition of the categorified algebraic K-theory groups of R

We need to impose some algebraic finiteness conditions on our A_∞ categories, otherwise any K-theory we construct would be trivial due to the Eilenberg swindle, the same reason that K-theory of infinite dimensional vector bundles is trivial. A natural requirement is that our A_∞ categories be smooth and proper, or saturated in the sense of Kontsevich-Soibelman [9], as this is a dualizability condition, see [22]. In the $\mathbb{Z}_n$-graded setting, the analogue of this condition is more complex, we will instead work with a less restrictive notion, which is not too hard to check for geometric examples.

Set $\mathcal{A} = \mathcal{A}_R \subset A_\infty Cat_R^{\mathbb{Z}_2, strict}$ to be the full subcategory whose objects are finite type, where the latter is defined as follows.

Definition 3.1. We say that $B \in A_\infty \operatorname{Cat}_R^{\mathbb{Z}_2, strict}$ is ***finite type*** if the following conditions are satisfied.

- B is Morita equivalent to an A_∞ algebra, whose underlying chain complex is projective and finitely generated over R in each degree.
- The Hochschild homology complex of B over R is quasi-isomorphic (in the category of 2-periodic complexes over R) to a complex that is projective and finitely generated over R in each degree.

Set $w\mathcal{A}_R \subset \mathcal{A}_R$ to be subcategory with the same objects and morphisms Morita equivalences.

Definition 3.2. Let X be a simplicial set. A ***bundle of*** A_∞ ***categories over*** X is a functor $F : \Delta(X) \to w\mathcal{A}_R$. F will be called a ***bundle functor over*** X.

Suppose we have bundle functors over X, fitting into a sequence of natural transformations of functors to $\mathcal{A}_R$ (as opposed to $w\mathcal{A}_R$):

$$F_0 \to F_1 \to F_2, \tag{3.3}$$

such that for each object $\Sigma \in \Delta(S^k)$ the resulting sequence in $\mathcal{A}_R$ is a short exact sequence. Then we call $F_0 \to F_1 \to F_2$ a ***short exact sequence***.

Definition 3.4. Let X be a Kan complex, we define $K^{Cat}(X, R)$ to be the quotient of the free abelian group generated by the set of concordance classes $[F]$ of functors $F : \Delta(X) \to w\mathcal{A}_R$, by the subgroup generated by

$$\{[F_0] - [F_1] + [F_2] \,|\, \text{there is a short exact sequence } F_0 \to F_1 \to F_2\}.$$

This is clearly contravariantly functorial in X so if $f : X \to Y$ is a simplicial map then we get a group homomorphism $f^* : K^{Cat}(Y, R) \to K^{Cat}(X, R)$ defined on generators by $f^*([F]) = [F \circ \Delta f]$.

As classical algebraic K-theory groups, $K^{Cat}(X, R)$ is covariantly functorial in R by "extension of scalars". Furthermore, it is a contravariant homotopy functor in X, as by definition of concordance it is clear that if f is homotopic to g then $f^* = g^*$. In particular if X_0 is homotopy equivalent to X_1 then $K^{Cat}(X_0, R) = K^{Cat}(X_1, R)$.

Let $S^n_\bullet$ be the singular complex of S^n.

Definition 3.5. For $k > 0$, we define

$$K^{Cat}_n(R) = \ker\{(\Delta^0 \to S^n_\bullet)^* : K^{Cat}(S^n_\bullet, R) \to K^{Cat}(\Delta^0, R)\},$$

and define

$$K_0^{Cat}(R) = K^{Cat}(\Delta^0, R).$$

Remark 3.6. The above definition of the groups $K_n^{Cat}(R)$ is meant to be very direct, and amenable to computation. However, for the sake of theory and also for certain computations it is important to demonstrate why these groups are in fact the Waldhausen K-theory groups. That is we want them to be homotopy groups of an infinite loop space $K^{Cat}(R)$, associated to a certain Waldhausen category underlying $\mathcal{A}_R$. The needed category is already described by Toën in [21]. We take the full subcategory $\widehat{\mathcal{A}}_R \subset \mathcal{A}_R$ with objects Morita fibrant (pretriangulated and idempotent complete). To be strict this will not be a Waldhausen category but ∞-Waldhausen in the sense of Barwick [1], as for example the zero category 0 (the category with one object and zero complex as endomorphism set) is not initial in $\widehat{\mathcal{A}}_R$ but the hom-sets $hom_{\widehat{\mathcal{A}}_R}(0, A)$ are contractible when taken in the ∞-localization of $\widehat{\mathcal{A}}_R$ with respect to the Morita equivalences. Then $K^{Cat}(R)$ is defined to be $K(\widehat{\mathcal{A}}_R)$ - the Waldhausen K-theory spectrum.

3.1. **A very basic computation.** We recall the definition of $K_0(R)$. This is the quotient of the free abelian group generated by the set of isomorphism classes of projective finitely generated R-modules, by the subgroup generated by

$$\{[A] - [B] + [C] \,|\, \text{exists an exact sequence } 0 \to A \to B \to C \to 0.\}$$

If $\mathcal{E}$ is a triangulated category, one defines $K_0(\mathcal{E})$ to be the quotient of the free abelian group, generated by the set of isomorphism classes in $\mathcal{E}$, by the subgroup generated by

$$\{[A] - [B] + [C] \,|\, \text{exists a distinguished triangle } A \to B \to C \to A[1]\}.$$

It is known that $K_k(D^{perf}(Mod_R)) \simeq K_k(R)$, where Mod_R denotes the category of projective finitely generated R-modules and $D^{perf}(Mod_R)$ denotes the derived category of perfect complexes.

If $D_n(Mod_R)$ denotes the n-periodic derived category, we no longer have that $K_0(R) \simeq K_0(D_n(Mod_R))$, as there are certain corrections, see Saito [15]. However we have the following.

Theorem 3.7 (Saito [15]). *For n even, the natural homomorphism:*

$$\psi : K_0(D_n(Mod_R)) \to K_0(R) \tag{3.8}$$

$$\psi([V]) = \sum_{i=0}^{i=n} (-1)^i [H^i(V)], \tag{3.9}$$

is an isomorphism.

There is a group homomorphism

$$\phi : K_0^{Cat}(R) \to K_0(D_2(Mod_R))$$

induced by $[C] \mapsto [Hoch_\bullet(C)]$ for $Hoch_\bullet(C)$ the Hochschild homology complex of C. This readily follows from the fact that the Hochschild functor $Hoch_\bullet$ is "localizing" which means in this case that a short exact sequence

$$C_0 \to C_1 \to C_2$$

in $\mathcal{A}_R$ is taken to an exact triangle

$$Hoch_\bullet(C_0)] \to Hoch_\bullet(C_1) \to Hoch_\bullet(C_2) \to Hoch_\bullet(C_0)[1],$$

in the derived category $D_2(Mod_R)$, see Keller [7, Theorem 5.2].

Corollary 3.10. *The composition map*

$$K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}})) \xrightarrow{\psi} K_0(\mathbb{Z}),$$

is non-trivial.

Proof. Take C to be a finite type $\mathbb{Z}_2$-graded A_∞ algebra over $\mathbb{Z}$ such that $HH_0(C) = \mathbb{Z}$, $HH_1(C) = 0$, $HH_2(C) = \mathbb{Z}$, where $HH_\bullet(C)$ is the homology of $Hoch_\bullet(C)$. As a geometric example take $C = FH_\bullet(L_0, L_0)$ where the latter is the Floer chain algebra of the equator L_0 in S^2, then the Hochschild homology $HH_\bullet(C)$ is isomorphic to the singular homology $H_\bullet(S^2, \mathbb{Z})$, [19].

So we get

$$\psi([Hoch_\bullet(C)]) = 2[\mathbb{Z}] \neq 0 \in K_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

□

A further corollary of the proof above is:

Corollary 3.11. $[\mathrm{Fuk}(S^2, \omega)]$ *represents a non-trivial element in* $K_0^{Cat}(\mathbb{Z})$.

One natural question is how to produce interesting elements in $K_m^{Cat}(R)$ for higher m. We will construct such elements using Hamiltonian fibrations and Floer theory.

4. Bundles of A_∞ categories from Hamiltonian fibrations

The main construction of [17] has input a smooth Hamiltonian fibration $M \hookrightarrow P \to Y$, with (M, ω) a compact monotone symplectic manifold, e.g. $(\mathbb{CP}^n, \omega_{FS})$. Denote by $Y_\bullet$ the simplicial set of smooth maps $\Delta^n \to Y$ for Δ^n also denoting the standard topological n-simplex.

The output is a functor

$$F_P : \Delta(Y_\bullet) \to wA_\infty Cat_R^{\mathbb{Z}_2, strict}, \tag{4.1}$$ [3]

that is a bundle functor over $Y_\bullet$, whose concordance class F_P is independent of auxiliary choices (these mainly consist of choices of almost complex structures and Hamiltonian connections.) For $x : \Delta^0 \to Y$ a point map $F_P(x) = \mathrm{Fuk}(P_x, \omega_x)$, P_x denoting the fiber over x.

We can outline this further, if $\Sigma : \Delta^1 \to Y$ is a path from x to y, then $F_P(\Sigma)$ has the set objects

$$\mathrm{obj}\, F_P(x) \sqcup \mathrm{obj}\, F_P(y).$$

For $L_x \in \mathrm{obj}\, F_P(x)$ and $L_y \in \mathrm{obj}\, F_P(y)$, $hom_{F_P(\Sigma)}(L_x, L_y)$ is a certain chain complex generated by flat sections of an auxiliary Hamiltonian connection on $\Sigma^* P$, with end points of the sections on L_x and L_y.

Remark 4.2. Thus $F_P(\Sigma)$ also gives a $(F_P(x), F_P(y))$ bimodule and a $(F_P(y), F_P(x))$ bimodule, which are mutual inverses in the Morita category $Hmo^{\mathbb{Z}_2}$, this gives another bridge to Morita theory.

More generally, for a smooth $\Sigma : \Delta^n \to Y$ denote by $v_i : \Delta^0 \to Y$, $0 \leq i \leq n$, the corresponding vertex maps. Then $F_P(\Sigma)$ has the set of objects

$$\sqcup_i\, \mathrm{obj}\, F_{P(\Sigma)}(v_i), \tag{4.3}$$

the hom sets are defined analogously so that an edge morphism $\Sigma^1 \to \Sigma$ in $\Delta(Y_\bullet)$ induces a fully-faithful embedding $F_P(\Sigma_1) \to F_P(\Sigma)$. The only deep part of the construction is then the definition of the A_∞ structure on $F_P(\Sigma)$. This involves some geometric topology of the universal curve, but we will not attempt to outline this further.

Definition 4.4. We call a compact monotone symplectic manifold ***finite monotone type***, if its monotone Fukaya category (see [19] and also [17]) is finite type.

We then have the following corollary of [17, Theorem 6.6].

[3] In [17] we work with $R = \mathbb{Q}$ but this was only used for inverting quasi equivalences. As we don't need to do this here, the construction works over any commutative ring.

Theorem 4.5. *Let $M \hookrightarrow P \to Y$ be a Hamiltonian fibration with fiber (M, ω) finite monotone type. Then the assignment:*

$$[P] \mapsto [F_P] \in K^{Cat}(Y_\bullet, R) \tag{4.6}$$

is well defined, where $[P]$ is the Hamiltonian isomorphism class of the fibration P.

As an example, we have:

Proposition 4.7. *$(\mathbb{CP}^n, \omega_{FS})$ is finite monotone type.*

Proof. Specifically, we take a concrete model of $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ with the set of objects given by the orbit of the Clifford torus L_0 by the natural $\mathrm{Ham}(\mathbb{CP}^n, \omega)$ action. In this case, it is automatic that $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ is quasi-isomorphic to the A_∞ algebra $hom_{\mathrm{Fuk}(\mathbb{CP}^n,\omega)}(L_0, L_0)$. The Hochschild chain complex of the latter is quasi-isomorphic to the Hamiltonian Floer chain complex, see for instance [19], which in each degree is finitely and freely generated over $\mathbb{Z}$. And so $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ is finite type for any R. $\square$

In particular, taking $Y = S^m$ and composing with the natural map

$$K^{Cat}(S^m_\bullet, R) \simeq K^{Cat}(S^m_{simp}, R) \to K^{Cat}_m(R)$$

we get:

Corollary 4.8. *For each commutative ring R, there are natural maps:*

$$\phi_m : \pi_m(B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})) \to K^{Cat}_m(R). \tag{4.9}$$

And in particular natural maps:

$$\phi_m : \pi_m(BPU(n)) \to K^{Cat}_m(R). \tag{4.10}$$

Proof. Let $\mathbb{CP}^n \hookrightarrow P \to B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$ be the tautological Hamiltonian fibration, that is the fiber bundle associated to the universal principal $\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$ bundle over $B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$. Use the conjunction of Proposition 4.7 and Theorem 4.5 to get the needed conclusion. $\square$

The maps ϕ_m are group homomorphisms. This immediately follows once it is shown that

$$K^{Cat}_m(R) = \pi_m(K^{Cat}(R)),$$

with $K^{Cat}(R)$ as in Remark 3.6. This is the domain of pure algebraic topology, there is a more general statement that holds for any Waldhausen category. The formal verification will not be carried out here, (it is also very likely already known.)

4.1. Generalizing to G/T.

Given a compact Lie group G, the maximal coadjoint orbit G/T can be made monotone with respect to the Kirillov-Kostant-Souriau symplectic structure ω_{KKS}, which for G/T is uniquely determined up to deformation equivalence, see for instance [8]. We will from now on omit this monotone ω_{KKS} from notation, but implicitly it will always be the chosen symplectic form. If $G = PU(n)$ there is a distinguished monotone Lagrangian torus L_{GC} of $PU(n)/T$, arising as the central fiber of a Gelfand-Cetlin system, see [3, Theorem B]. This is a kind of analogue of the Clifford torus.

It is natural to expect that $Fuk(G/T)$ is finite type. In this case we similarly get homomorphisms $\pi_m(B\,\mathrm{Ham}(G/T) \to K^{Cat}_m(R)$. Note that when we talk about $Fuk(G/T)$, we are talking about a model [4] with finitely many objects up to the $\mathrm{Ham}(G/T)$ action, similarly to how we treated $Fuk(\mathbb{CP}^n)$.

[4] Not necessarily Morita equivalent to $Fuk(G/T)$.

For a concrete example, form the sub-category $C_{L_{GC}}$ of $Fuk(PU(n)/T)$ with objects formed by the $\mathrm{Ham}(PU(n)/T)$-orbit of L_{GC}. In this case the finite type condition is forced and we get natural homomorphisms

$$\phi_{L_{GC},m} : \pi_m(B\,\mathrm{Ham}(PU(n)/T)) \to K_m^{Cat}(R),$$

by the same construction; that is getting bundles of A_∞ categories over S^m with fiber $C_{L_{GC}}$. More explicitly, given a fibration $PU(n)/T \hookrightarrow P \to S^m$, and $x : \Delta^0 \to Y$ a point map, set $F_P^{\mathcal{L}_{GC}}(x) = C_{\mathcal{L},x}$ where $C_{\mathcal{L}_{\mathcal{GC}}} \subset \mathrm{Fuk}(P_x, \omega)$, (identifying P_x with $PU(n)/T$ up to $\mathrm{Ham}(PU(n)/T)$ action). For a general Σ, $F_P^{\mathcal{L}_{GC}}(\Sigma)$ is a category with finitely many objects, all of which are isomorphic to a distinguished object, corresponding to $\mathcal{L}_{GC}$; this follows by the prescription in (4.3).

If we restrict to $G = PU(n)$, we get two maps $\phi_{L_{GC},m} : \pi_m(BPU(n)) \to K_m^{Cat}(R)$, and $\phi_m : \pi_m(BPU(n)) \to K_m^{Cat}(R)$. These are likely distinct, they correspond to taking different coadjoint orbits of G, and so different fibrations over $BPU(n)$.

We can also do this for G/T (and more general coadjoint orbits) by constructing a distinguished Lagrangian $\mathcal{L}$, as the central fiber of a toric degeneration of G/T. One of the most well known setups for this is via so called string polytopes and Newton–Okounkov bodies see [4], and G/T always has a toric degeneration. For a general G it is expected that $\mathcal{L}$ is monotone, (this in principle depends on the chosen degeneration, but it is enough for us that exists one degeneration, such that $\mathcal{L}$ is monotone). For example, this holds for $PU(n)/T$ as in this case $\mathcal{L}$ can be identified with L_{GC} for the toric degeneration associated to the Gelfand–Cetlin polytope.

Assuming this for the moment we get a finite type sub-category category $C_{\mathcal{L}} \subset \mathrm{Fuk}(G/T)$, with the set of objects the orbit of $\mathcal{L}$ by the $\mathrm{Ham}(G/T)$ action. If we take this $C_{\mathcal{L}}$ as our model we again get maps:

$$\phi_{\mathcal{L}} : \pi_m(B\,\mathrm{Ham}(G/T)) \to K_m^{Cat}(R). \tag{4.11}$$

5. Hamiltonian elements in algebraic K-theory

One of the goals is to obtain "Hamiltonian elements" in classical algebraic K-theory of R. Let $M \hookrightarrow P \to Y$ be a Hamiltonian fibration, with compact monotone fiber (M, ω). Let

$$F_P : \Delta(Y_\bullet) \to wA_\infty Cat_R^{\mathbb{Z}_2, strict}$$

be as in the previous section. Then the composition: $\Phi_P = Hoch_\bullet \circ F_P$ gives a functor to $wCh_R^{\mathbb{Z}_2}$, where $Ch_R^{\mathbb{Z}_2}$ is the Waldhausen (strict) category $Ch_R^{\mathbb{Z}_2}$ of 2-periodic complexes of projective finite rank R-modules. Its Waldhausen K-theory infinite loop space will be denoted by $K^{\mathbb{Z}_2}(R)$.

Restrict again to $Y = B\,\mathrm{Ham}(G/T)$, and P the associated G/T bundle over $\mathrm{BHam}(G/T)$. Now by standard homotopy theory, $|\Delta(Y_\bullet)| \simeq X$, where for a category C, $|C|$ is shorthand for the geometric realization of the nerve. Then we get a map:

$$|F_P| : B\,\mathrm{Ham}(G/T) \to |wCh_R^{\mathbb{Z}_2}|,$$

whose homotopy class is well defined by the fact that the concordance class of F_P is well defined.

There is a natural map $|wCh_R^{\mathbb{Z}_2}| \to K^{\mathbb{Z}_2}(R)$, by generalities of the Waldhausen construction, see [23, Remark 8.3.2]. Composing with this map we get a natural homotopy class:

$$[|F_P| : B\,\mathrm{Ham}(G/T) \to K^{\mathbb{Z}_2}(R)]. \tag{5.1}$$

Proof of Theorem 1.4. Compose the natural map

$$BG \to \mathrm{BHam}(G/T),$$

and the map from (5.1) to obtain h. ∎

6. Langlands dual algebraic K-theory elements.

It is natural to ask how Langlands duality fits into our story. Denote by $K^\vee$ the maximal compact subgroup of the Langlands dual $G^\vee_\mathbb{C}$ of the complexification $G_\mathbb{C}$. Via Theorem 1.4, we have natural maps:

$$h_{G,m} : \pi_m(BG) \to K^{\mathbb{Z}_2}_m(R),$$
$$h_{K^\vee,m} : \pi_m(BK^\vee) \to K^{\mathbb{Z}_2}_m(R).$$

If G is simply laced (e.g. $G = SU(n)$) it is not hard to see that the images of these maps are the same. In this case $G/T = K^\vee/T$ and then the images of G and $K^\vee$ in $\mathrm{Ham}(G/T)$ coincide, so that the assertion readily follows. So we may propose a conjecture :

Conjecture 1. *For each compact Lie group G the images of $h_{G,m}$ and $h_{K^\vee,m}$ coincide.*

Furthermore, there should be "lifts" as in (4.11):

$$\phi_{\mathcal{L},G} : \pi_m(BG) \to K^{Cat}_m(R),$$
$$\phi_{\mathcal{L},K^\vee} : \pi_m(BK^\vee) \to K^{Cat}_m(R),$$

and we can again ask if the images coincide. For $m = 0$, this is saying that $Fuk(G/T)$ and $Fuk(K^\vee/T)$ [5] represent the same classes in $K^{Cat}_0(R)$. This assertion holds if these categories were Morita equivalent, but it is much weaker. The case $m = 0$ is then comparable to the homological mirror symmetry conjecture of Kontsevich.

The above is thematically connected to topological aspects of Langlands duality/correspondence as in [2] for example. But I don't know a direct mathematical connection. From a physical perspective of Kapustin-Witten [5], geometric/topological aspects of Langlands correspondence are expressions of the electric-magnetic duality. It would be good to understand if there are likewise some physical connotations of the maps $h_{G,m}$.

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